

- 381.** $A + B + C + D = 47$... (i)
 $A + B = 27$... (ii)
 $A + C = 25$... (iii)
 $A + D = 23$... (iv)
 Adding (ii), (iii) and (iv), we get:
 $2A + (A + B + C + D) = 75 \Leftrightarrow 2A + 47 = 75 \Leftrightarrow 2A = 28 \Leftrightarrow A = 14$.
 Putting $A = 14$ in (ii), we get: $B = 13$.
- 382.** Let the sum be ₹ x .
 Then, $\frac{x}{12} - \frac{x}{16} = 400 \Leftrightarrow \frac{x}{48} = 400 \Leftrightarrow x = 400 \times 48 = 19200$.
- 383.** Let the total number of sweets be $(25x + 8)$.
 Then, $(25x + 8) - 22$ is divisible by 28
 $\Leftrightarrow (25x - 14)$ is divisible by 28
 $\Leftrightarrow 28x - (3x + 14)$ is divisible by 28
 $\Leftrightarrow (3x + 14)$ is divisible by 28 $\Leftrightarrow x = 14$.
 $\therefore$ Total number of sweets = $(25 \times 14 + 8) = 358$.
- 384.** Let the total number of sweets be x . Then,
 $\frac{x}{80} - \frac{x}{112} = 6 \Leftrightarrow \frac{7x - 5x}{560} = 6 \Leftrightarrow \frac{x}{280} = 6 \Leftrightarrow x = 1680$.
 $\therefore$ Number of sweets each child was originally supposed to get = $\left(\frac{1680}{112}\right) = 15$.
- 385.** Let the total number of children be n and the amount received by each child be ₹ x .
 Then, total sum distributed = ₹ (nx) .
 So, $(n + 16)(x - 2) = nx$
 $\Leftrightarrow nx - 2n + 16x - 32 = nx$
 $\Leftrightarrow 16x - 2n = 32$... (i)
 And, $(n - 16)(x + 3) = nx$
 $\Leftrightarrow nx + 3n - 16x - 48 = nx$
 $\Leftrightarrow 16x - 3n = -48$... (ii)
 Subtracting (ii) from (i), we get: $n = 80$.
 Putting $n = 80$ in (i), we get: $16x = 192$ or $x = 12$.
 Hence, total sum distributed = $nx = ₹ (80 \times 12) = ₹ 960$.
- 386.** Suppose, Sanya and Vidushi donate money to x and $(x + 5)$ people respectively.
 Then, $\frac{100}{x} - \frac{100}{x+5} = 1 \Leftrightarrow 100(x + 5) - 100x = x(x + 5)$
 $\Leftrightarrow x^2 + 5x - 500 = 0$
 $\Leftrightarrow (x - 20)(x + 25) = 0 \Leftrightarrow x = 20$.
 $\therefore$ Total number of recipients of charity = $x + (x + 5) = (2x + 5) = 45$.
- 387.** Let the number of girls be x . Then, number of boys = $3x$.
 Total number of students = $x + 3x = 4x$.
 Clearly, the total number of students must be a multiple of 4.
- 388.** Let the number of boys be x . Then, number of girls = $(x - 2)$.
 $\therefore x^2 - (x - 2)^2 = 28$
 $\Leftrightarrow x^2 - (x^2 + 4 - 4x) = 28$
 $\Leftrightarrow 4x - 4 = 28$
 $\Leftrightarrow 4x = 32$
 $\Leftrightarrow x = 8$.
 So, number of boys = 8; number of girls = 6.
 $\therefore$ Total number of boys and girls = $8 + 6 = 14$.

- 389.** Let number of boys = x . Then, number of girls = x .
 Now, $2(x - 8) = x$ or $x = 16$.
 $\therefore$ Total number of students = $2x = (2 \times 16) = 32$.
- 390.** Given exp. = $(1 + x)(1 - x)(1 + x^2)(1 + x^4)(1 + x^8)$
 $= (1 - x^2)(1 + x^2)(1 + x^4)(1 + x^8) = (1 - x^4)(1 + x^4)(1 + x^8)$
 $= (1 - x^8)(1 + x^8) = (1 - x^{16})$.
- 391.** Given exp. = $\left(\frac{1}{x-1} - \frac{1}{x+1}\right) - \frac{2}{x^2+1} - \frac{4}{x^4+1}$
 $= \left[\frac{(x+1) - (x-1)}{(x-1)(x+1)}\right] - \frac{2}{x^2+1} - \frac{4}{x^4+1}$
 $= \left(\frac{2}{x^2-1} - \frac{2}{x^2+1}\right) - \frac{4}{x^4+1}$
 $= \left[\frac{2(x^2+1) - 2(x^2-1)}{(x^2-1)(x^2+1)}\right] - \frac{4}{x^4+1}$
 $= \frac{4}{x^4-1} - \frac{4}{x^4+1} = \frac{4(x^4+1) - 4(x^4-1)}{(x^4-1)(x^4+1)}$
 $= \frac{8}{(x^8-1)}$.
- 392.** Given exp. = $\left[\left(x + \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} - x \cdot \frac{1}{x}\right)\right] \left[\left(x - \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} + x \cdot \frac{1}{x}\right)\right]$
 $= \left(x^3 + \frac{1}{x^3}\right)\left(x^3 - \frac{1}{x^3}\right) = \left(x^6 - \frac{1}{x^6}\right)$.
- 393.** $x + \frac{1}{x} = 3 \Rightarrow \left(x + \frac{1}{x}\right)^2 = 3^2 \Rightarrow x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x} = 9$
 $\Rightarrow x^2 + \frac{1}{x^2} = 9 - 2 = 7$.
- 394.** $a - \frac{1}{a} = p \Rightarrow \left(a - \frac{1}{a}\right)^2 = p^2 \Rightarrow a^2 + \frac{1}{a^2} - 2 \cdot a \cdot \frac{1}{a} = p^2$
 $\Rightarrow a^2 + \frac{1}{a^2} = p^2 + 2$.
- 395.** $\frac{x-1}{x} = 3 \Rightarrow 1 - \frac{1}{x} = 3 \Rightarrow \left(1 - \frac{1}{x}\right)^2 = 9$
 $\Rightarrow 1 + \frac{1}{x^2} - \frac{2}{x} = 9$
 $\Rightarrow 1 + \frac{1}{x^2} = 9 + \frac{2}{x} = 9 + \frac{2}{\left(-\frac{1}{2}\right)} = 9 - 4 = 5$.
 $\left[\because \frac{x-1}{x} = 3 \Rightarrow 3x = x-1 \Rightarrow 2x = -1 \Rightarrow x = -\frac{1}{2}\right]$
- 396.** $x^2 + \frac{1}{x^2} = 34 \Rightarrow x^2 + \frac{1}{x^2} + 2 \cdot x^2 \cdot \frac{1}{x^2} = 34 + 2 = 36$
 $\Rightarrow \left(x + \frac{1}{x}\right)^2 = 36 \Rightarrow \left(x + \frac{1}{x}\right) = 6$.